Md. Aber Islam

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Education

BRAC University Dhaka, Bangladesh
Bachelor of Science in Computer Science and Engineering Jul 2020 – Jan 2025

Experience

Nybsys Inc Dhaka, Bangladesh
AI/ML Engineer Apr 2026 – Present

- Developed agentic systems for Grameenphone RAN engineers as an embedded Nybsys resource, analyzing telecom data to support network-health assessment, alarm investigation, and KPI analysis.
- Built Qwen agent workflows with LangChain, LangGraph, anomaly detection, custom MCP tools via FastMCP, and Langfuse observability; deployed private LLM inference services on Kubernetes.
- Developed a quality-control (QC) station for T-shirt print inspection, combining image processing and anomaly detection with OpenCV, GStreamer, and YOLO; used MLflow for experiment tracking.

RACO AI Dhaka, Bangladesh
AI Engineer Dec 2025 – Apr 2026

- Built an end-to-end call-system pipeline integrating automatic speech recognition (ASR) and text-to-speech (TTS), with a focus on Bangla speech models and noise reduction.
- Fine-tuned LLMs and built ML pipelines for training, evaluation, and deployment, integrating Bangla ASR/TTS models and optimizing inference workflows.
- Built a social-media automation engine supporting posting and messaging workflows across LinkedIn, Twitter/X, Facebook, and Messenger.

Join Venture AI Dhaka, Bangladesh
AI Developer Oct 2025 – Dec 2025

- Delivered RAG systems and agent workflows for multiple startup products, automating business processes and reducing manual work by 60%.
- Built production APIs and NLP pipelines using FastAPI, vector search, LangChain, and Hugging Face Transformers.
- Implemented testing and monitoring for deployed AI features, maintaining 99% uptime.

Aspire to Innovate (a2i) Program Dhaka, Bangladesh
Machine Learning Intern Jun 2024 – Dec 2024

- Built Bangla text-classification pipelines achieving 92% accuracy and developed web scrapers for large-scale Bangla NLP corpus collection.
- Automated data preprocessing workflows, reducing processing time by 50%.
- Processed 100,000+ public-service records and implemented ML models to support evidence-based policy analysis.

Skills

Languages: Python, SQL, Go, JavaScript
ML/NLP: PyTorch, Hugging Face Transformers, PEFT/LoRA, scikit-learn, NumPy, Pandas, STT/TTS
GenAI/RAG: LLM fine-tuning, RAG, FAISS, LangChain, LangGraph, Qwen, MCP, FastMCP
Computer Vision: OpenCV, GStreamer, YOLO, image processing, defect detection
Evaluation/Monitoring: MLflow, Langfuse, anomaly detection
Backend/Infrastructure: FastAPI, Pydantic, Docker, Kubernetes, local LLM inference, model serving
Tools/UI: Git, GitHub, Streamlit, Gradio
Domain: RAN health, alarms and KPIs; garment-print inspection; Bangla NLP

Selected Projects

banglatok — Bangla/Banglish Tokenizers | *GitHub* | *PyPI*

- Published a typed Python API backed by a native Go Unigram/BPE engine for Bangla, English, and Banglish with zero Python runtime dependencies.
- Trained models on 563,476 records; shipped offline models, a CLI, and Hugging Face exports preserving token IDs.
- Verified lossless normalized round trips on 10,000 unseen BanglishRev reviews; measured 1.1–1.7x faster single-threaded encoding than same-data, same-size SentencePiece Unigram on Apple M5 and Linux x86_64.

Medical Guideline RAG System | *GitHub* | *Live Demo*

- Built a medical-guideline research assistant with Gemini and NetworkX for medication guidance and drug-interaction analysis across 33 medical domains.
- Indexed 50,000+ medical articles with FAISS, achieving <200 ms search latency.
- Delivered a FastAPI backend and Streamlit interface supporting 100+ concurrent consultations.

Mistral-7B Fine-tuning for Medical QA | *GitHub* | *Paper*

- Fine-tuned Mistral-7B-Instruct on PubMedQA with PEFT/LoRA, training only 0.58% of model parameters (41.9M of 7.3B).
- Built PubMedQA and MMLU evaluation pipelines; achieved 73.0% accuracy and a 70.23% F1 score.
- Used bfloat16 mixed-precision training to reduce training time by 40%.

Real-time Face Mask Detection | *GitHub* | *Live Demo*

- Built a real-time face-mask detector using PyTorch, MobileNetV3, and transfer learning.
- Trained on 12,000+ balanced images; achieved 96% test accuracy across varied devices and lighting conditions.
- Created a Gradio interface supporting webcam, video, and image inputs.